GWENDAL LE COGUIC

COURSES

Each training session focuses on practical demonstrations, interactive exercises, and discussions based on real-life situations to ensure the acquisition of key skills.

Each session can be taken:

- on-site or remotely
- in French or English
- individually or in groups

Quotes available upon request

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1. HACKING & WEB APPLICATION SECURITY



a. Description

*This intensive 3-day training is designed for **developers** and **system administrators** aiming to enhance the security of their web applications. Combining **theoretical knowledge** with **practical exercises**, the course covers **common vulnerabilities**, utilization of **essential tools**, **best practices** in secure web development. Participants will learn to **identify**, **exploit**, and **remediate** security flaws, gaining hands-on experience to proactively defend against real-world threats.*

b. Educational goals

- Understand the main web application vulnerabilities
- Learn how to detect and exploit common security flaws
- Use tools like *Burp Suite* and *SQLmap* effectively
- Apply secure coding practices to prevent attacks
- Improve overall application resilience against threats

c. Prerequisites

- Basic knowledge of web development (HTML, CSS, JavaScript)
- Familiarity with web protocols (HTTP, HTTPS)
- Experience with using command-line tools

d. Targeted audience

- Web developers
- System administrators
- Dev(Sec)Ops engineers
- Any cybersecurity enthusiasts

e. Program

Day 1 - The basics of web security

The landscape of web insecurity

- Web security: a vast topic
- Architecture of a typical Web app
- Open Web Application Security Project (OWASP)
- CWEs, CVEs & CVSS
- External and internal attackers
- The different steps of an attack
- An attacker's bag of tricks
- Common misconceptions

Security's place in the development cycle

- The importance of inventory
- Shadow IT
- Production bias & incompatibility with Agile approaches
- The idea of *DevSecOps*
- The importance of testing & code reviews
- Static & dynamic analysis

Outside help

- Security audits and *Bug Bounty*
- Vulnerability disclosure policy

HTTP protocol

- Reminder about the HTTP(s) protocol
- Anatomy of HTTP requests & responses
- Semantics & status codes
- Introduction to *Dev Tools*
- Introduction to *Burp Suite*

URLs

- Anatomy of a URL
- Open redirects exploitation

Disclosure of sensitive information

- Sequential IDs
- Verbose error messages
- Directory listing
- Clear-text logging of sensitive information
- Secrets on the client side
- API Excessive data exposure
- Public *Git* repositories & history
- *Git* directory deployed

Day 2 - User input flaws

Input validation

- Client-side, server-side, or both?
- HTTP parameter pollution
- Mass assignment

Cross-Site Scripting (XSS)

- Cross-site scripting is everywhere
- Types of cross-site scripting
- What is the impact?
- How to protect & how to bypass

SQL Injection (SQLi)

- What is it?
- What is the impact?
- How to protect & how to bypass
- Presentation of *SQLMap*
- *NoSQL* injection

Content Security Policy (CSP)

- Definition
- Most common directives
- Possible configuration
- What is the impact?
- How to protect & how to bypass

Other forms of injection

- HTML injection
- Code injection
- OS command injection

Files

- Common file issues
- Arbitrary file upload
- Path traversal
- Local & remote file inclusion

Day 3 - Authenticated user vulnerabilities

Authentication & vulnerabilities

- Authentication
- User enumeration
- Rate limiting and security questions
- Multiple-factor authentication

Passwords

- Cryptography in a nutshell: encoding, hashing, encryption
- Password-strength policy
- Leaked passwords
- Password storage
- Cracking tools

Cookies

- Cookies attributes: *secure, httponly, domain, samesite…*
- Session hijacking

Cross-Site Request Forgery (CSRF)

- What is it?
- What is the impact?
- How to protect & how to bypass

Clickjacking

- What is it?
- What is the impact?
- How to protect & how to bypass

Authorization & access control

- Authentication vs. authorization
- Roles and permissions
- Insecure direct object reference

Cross-Origin Resource Sharing (CORS)

- Refresher on the *Same-Origin Policy*
- Possible misconfigurations
- What is the impact?
- How to protect

2. ADVANCED HACKING & WEB APPLICATION SECURITY



a. Description

*An intensive 2-day training designed for professionals with a solid foundation in web security. It delves into advanced vulnerabilities and equips participants with the skills to **identify**, **exploit**, and **mitigate complex security issues** in web applications.*

b. Educational goals

- Identify and exploit advanced web vulnerabilities
- Understand complex attack vectors like SSRF, XXE, and prototype pollution
- Evaluate risks linked to third-party components and infrastructure
- Apply secure coding practices to mitigate real-world threats

c. Prerequisites

- Basic knowledge of web development (HTML, CSS, JavaScript)
- Familiarity with web protocols (HTTP, HTTPS)
- Experience with using command-line tools
- Familiar with basic web vulnerabilities (XSS, CSRF, SQLi…)

d. Targeted audience

- Web developers
- System administrators
- Dev(Sec)Ops engineers
- Any cybersecurity enthusiasts

e. Program

Day 1

- Carriage Return Line Feed injection (CRLF)
- Second order bugs
- Prototype pollution
- *PostMessage*
- Server-Side Request Forgery (SSRF)

Day 2

- Dependency confusion
- Subdomain takeover
- JSON Web Token security (JWT)
- XML External Entity injection (XXE)
- Emails security

3. CYBERSECURITY AWARENESS



a. Description

*This one-day awareness course is designed for all **employees and collaborators** who interact with digital tools in their daily work. The goal is to demystify cybersecurity, present the **major threats and risks** facing organizations today, and empower participants with concrete, practical knowledge to reduce vulnerabilities. Through clear **explanations**, **real-world examples**, and **actionable advice**, this session helps create a culture of shared responsibility around information security.*

b. Educational goals

- Understand the main cybersecurity threats and their impact
- Identify common attack techniques
- Recognize the importance of user behavior in preventing incidents
- Apply basic best practices in daily work
- React appropriately to potential security incidents
- Understand legal and organizational responsibilities

c. Prerequisites

None.

d. Targeted audience

- Employees at all levels: managers and team leaders, human resources, administrative staff…
- IT Support and technical teams
- Remote Workers or telecommuters

e. Program

Introduction to cybersecurity

- What is cybersecurity and why it matters
- Real-world examples of threats
- Importance of user behavior

Organizational roles & responsibilities

- Who does what in cybersecurity
- Shared responsibilities and interdepartmental cooperation
- Risk awareness and internal communication

Legal and regulatory context

- Overview of GDPR and other key regulations
- What the law says about data protection and incident response
- Individual obligations and corporate compliance

Threat landscape & attack scenarios

- Common cyber threats
- Case studies and real incidents
- Understanding the human factor in cyberattacks

Best practices for secure behavior

- Password hygiene and multi-factor authentication
- Email security and identifying phishing attempts
- Safe browsing and social media usage

Securing digital tools & environments

- Safe use of devices (computers, phones, USBs)
- Working remotely securely
- Data backup and software updates

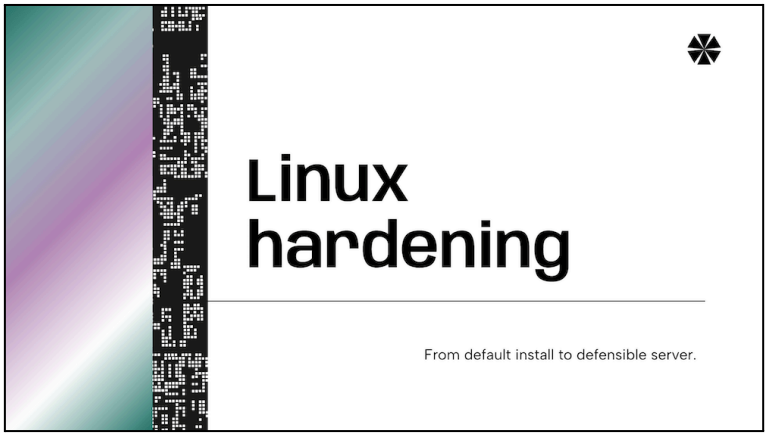
Responding to a cyber incident

- How to detect and report suspicious activity
- Immediate steps to take during a security breach
- Building a personal action plan

Wrap-up & recommendations

- Summary of key takeaways
- Resources for staying informed
- Open discussion and Q&A

4. LINUX HARDENING



a. Description

This intensive 1-day training teaches you how to harden a Linux server from the ground up. You'll learn to **minimize the system footprint**, **lock down user privileges**, **secure network exposure**, **and reinforce critical services**. Through hands-on exercises, you'll apply **real-world hardening** techniques used by security professionals.

b. Educational goals

- Identify common entry points, privilege-escalation paths, and misconfigurations
- Minimize installed software and apply kernel and filesystem protections
- Configure strong authentication, secure sudo policies, and controlled user environments
- Secure network interfaces, firewall rules, and remote access pathways
- Deploy logging, auditing, and basic intrusion-detection mechanisms

c. Prerequisites

- Being comfortable with CLI mode
- Familiarity with basic Linux administration

d. Targeted audience

- System administrators
- Dev(Sec)Ops engineers

e. Program

Introduction

- Why hardening matters?
- Understanding the attack surface
- Defense in depth
- Principle of least privilege
- Minimal install philosophy
- Hardening is role-specific
- Reference frameworks
- Baseline vs. hardening vs. compliance

System baseline

- Partitioning strategy
- Unnecessary packages & services
- Unused kernel modules
- Hardening the Kernel

Account hardening

- Mandatory Access Control (MAC)
- Account and sudo hardening
- Password and authentication policy

Network hardening

- Network exposure
- Firewalling
- Brute-force mitigation

Services hardening

- Dedicated users
- File permissions
- Service sandboxing

Filesystem, permissions & integrity

- SUID/SGID auditing
- Sticky bit and ACLs
- Immutable attributes
- File integrity monitoring

Logging, auditing & monitoring

- Kernel-level security event
- Centralized logging
- Intrusion detection concepts

Automation & compliance validation

- Scoring with Lynis
- Formal compliance scoring with OpenSCAP
- Hardening-as-code with Ansible

Resources

- Books
- Interactive platforms
- Courses & videos
- Tools

5. INTRODUCTION TO PENTESTING



a. Description

*The Pentest training is an intensive 3-day program designed to introduce **developers**, **system administrators** and any **cybersecurity enthusiasts** to the fundamentals of **ethical hacking**. Through a combination of **theory** and **practical exercises** in dedicated environments, participants will learn how to **identify and exploit vulnerabilities** in web applications and network services. Essential tools such as **Kali Linux**, **Metasploit**, **Burp Suite** and more will be covered. At the end of the training, participants will be able to conduct effective security audits and strengthen the protection of their IT infrastructures.*

b. Educational goals

- Understand the fundamentals of ethical hacking and legal boundaries
- Identify and analyze common vulnerabilities in web and network systems
- Use industry-standard tools like *Kali Linux, Metasploit,* and *Burp Suite*
- Perform manual exploitation techniques
- Conduct vulnerability assessments and interpret scan results
- Apply pentesting methodologies to simulate real-world attacks

c. Prerequisites

- Basic knowledge of IT systems and networks
- Familiarity with operating systems (Linux and Windows)
- Basic understanding of programming or scripting is a plus but not required

d. Targeted audience

- Developers
- System administrators
- Dev(Sec)Ops engineers
- Any cybersecurity enthusiasts

e. Program

Day 1 - Introduction and information gathering

Audit charter

- What is it?
- Content of the audit charter

Methodology

- Steps of an attack
- An attacker’s bag of tricks
- Testing process

Introduction to Kali Linux

- What is it?
- Tools included
- Lab configuration

Information gathering

- Passive vs. active recon
- *Google, Shodan, GitHub* hacking
- Collect Companies and employees data
- Technologies
- ASN enumeration
- DNS enumeration, resolution and probing
- Port scan

Vulnerability scan

- CWEs, CVEs & CVSS
- Scanners

Day 2 - Exploitation techniques and web security

Report

- What is it?
- Content of the report

HTTP testing

- Requirements
- What would you test?
- Browser extensions
- Introduction to *Burp Suite*
- Cross-Site Scripting
- SQL injection
- Arbitrary file upload

Other network services

- Shell vs. reverse shell
- Find, install and use exploits
- Introduction to *Metasploit*
- Service exploitation: *FTP, SMB, PostgreSQL, WebDAV…*

Day 3 - Post-exploitation and password attacks

Final presentation

- What is it?

Password attacks

- Cryptography refresher: encoding, hashing, encryption
- Leaked passwords and default passwords
- Online services & network attacks
- Wordlists
- Linux vs. Windows

Privilege escalation

- Horizontal vs. Vertical
- Linux vs. Windows

Post-exploitation

- Persistency
- Antivirus evasion
- Port forwarding and port tunneling

5. SÉCURITÉ LARAVEL



a. Description

*This one-day course is designed for **Laravel developers** who want to build secure applications from the ground up. You'll learn to **identify common vulnerabilities**, **apply Laravel's built-in protections**, and **adopt best practices** to secure authentication, **authorization**, **input validation**, and more.*

b. Educational goals

- Recognize common web vulnerabilities and Laravel's protections
- Implement secure authentication and authorization with Laravel tools
- Configure CORS and CSRF protections properly
- Validate and sanitize user input securely
- Handle file uploads and storage with security in mind

c. Prerequisites

- Good experience with PHP and Laravel framework
- Understanding of web application security concepts
- Familiarity with basic web vulnerabilities (XSS, CSRF, SQLi…)

d. Targeted audience

- Laravel developers
- Security professionals
- Dev(Sec)Ops engineers

e. Program

Global configuration

- Basic PHP security

- Global configuration
- Sessions
- *APP_KEY*

The basics
- Raw PHP in views
- Inputs validation
- Logging and Error handling
- Updates and Backups
- Tricking bots

Cookies & sessions
- Global configuration
- Laravel specific
- Session fixation

Passwords & encryption
- Encrypt/decrypt data
- Passwords validation
- Reset passwords

Authentication & permissions
- Official packages
- Laravel passport
- Gates and policies
- Users / roles / permissions

Rate limiting
- Usage
- Ratelimit actions / routes

File uploads & file storage
- What is it?
- Validation rules and Sanitization
- Storages configuration and usage
- Authorization / permissions / visibility

Mass assignment
- What is it?
- Expectation vs. reality
- How to protect?

Excessive data exposure
- What is it?
- How to protect?
- Take away

SQL Injection (SQLi)

- What is it?
- Prepared statements
- Parameter binding
- Dangers
- Take away

Cross-Site Scripting (XSS)

- What is it?
- Controllers
- Templates
- Input sanitization
- External helps

Cross-Site Request Forgery (CSRF)

- What is it?
- CSRF token
- Middleware
- Forms
- *X-csrf-token* and *X-xsrf-token*

Cross-Origin Resources Sharing (CORS)

- What is it?
- Configuration
- *allowed_origins* and *supports_credentials*

6. BUG BOUNTY OVERVIEW



a. Description

This half-a-day course offers a complete introduction to the world of bug bounty. ***Tailored for directors****,* ***CISO****,* ***team leader****, this session covers everything from the history and structure of bug bounty programs to real-world vulnerability reporting and triaging. Through* ***case studies****,* ***best practices****, and* ***industry insights****, participants will leave equipped with a strong foundation to begin (or improve) their bug bounty journey,* ***both from the hacker and company perspectives****.*

b. Educational goals

- Understand what a *bug bounty* program is
- Learn how *bug bounty* fits into a security strategy
- Identify key roles and responsibilities
- Know how to handle vulnerability reports
- Evaluate your organization's *bug bounty* readiness

c. Prerequisites

- Basic understanding of cybersecurity principles
- Familiarity with vulnerability management and risk assessment

d. Targeted audience

- Cybersecurity managers and security teams
- CISO and risk management professionals
- Organizations new to *bug bounty*

e. Program

Welcome & icebreaker

- Instructor intro & participant backgrounds
- What to expect from the course
- What is *bug bounty* and why it matters?

Understanding *bug bounty*: the ecosystem

- History and evolution of programs
- Roles and motivations: companies, platforms, hackers, 3rd parties
- *Pentest* vs. *bug bounty*: key differences

Anatomy of a *bug bounty* program

- How programs are structured
- Public vs. private programs
- What companies expect vs. what hackers expect

Reports: triaging & managing conflicts

- Lifecycle of a report
- Handling duplicates and disputes

Policy & program management

- What companies should prepare before launching
- Program evolution: scope, rewards, privacy
- Common mistakes to avoid

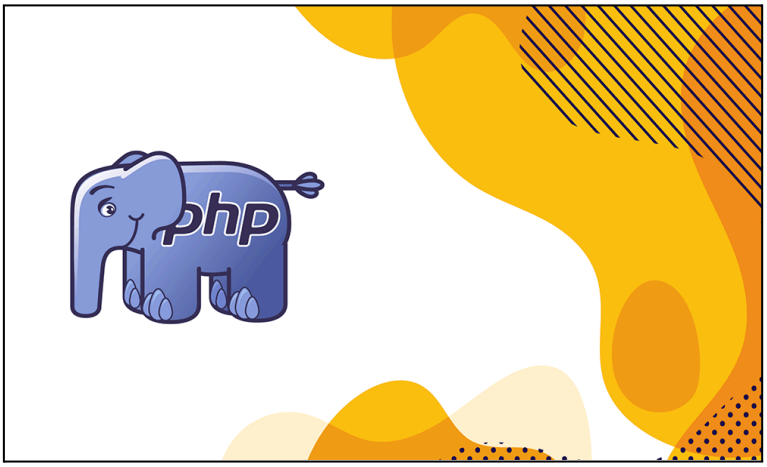
Rules, ethics & real case studies

- The unofficial rules of *bug bounty*
- Safe harbor, disclosure policies
- Failures and successes: *Verizon, Shopify, Zomato…*

Wrap-up & Q&A

- Recap of the day
- Open Q&A and personalized advice

7. PHP FOR BEGINNERS



a. Description

*A comprehensive 3-day training designed for beginners aiming to **build dynamic web applications using PHP**. Participants will learn to write structured PHP code, handle data with SQL, and integrate external libraries using Composer. The course emphasizes **practical learning through progressive exercises**, culminating in the development of a functional web application. By the end, attendees will be equipped to design and develop the server-side of a website, generating the necessary code for the front-end. This training is **ideal for novice developers** seeking to create applications or services based on the PHP programming language.*

b. Educational goals

- Understand PHP syntax and basic programming concepts
- Build dynamic web pages using PHP and HTML
- Manage data with MySQL and SQL queries
- Handle forms and user input securely
- Structure code using functions
- Use Composer to manage PHP libraries and dependencies

c. Prerequisites

- Basic knowledge of HTML and CSS
- Familiarity with web browsers and text editors

d. Targeted audience

- Beginners in web development
- Aspiring backend developers

- Anyone interested in PHP/MySQL to build dynamic websites

e. Program

Day 1 - PHP basics

What is PHP?

- A bit of History
- What is PHP and what is a PHP File?

PHP basics

- PHP tags & short tags
- PHP + HTML
- Comments

Variables

- Variables types
- Constants
- Quotes
- Concatenation
- Operators

Control Structures

- *if, elseif, else, while, for, while, for, foreach*
- *break, continue*
- *switch…case*
- *match*
- *include, require*

Functions

- Recursion
- Types declaration
- Variable arguments
- Default parameters
- Anonymous functions
- Arrow functions

HTTP methods

- Superglobals
- *GET* and *POST*
- Other HTTP methods
- Forms and filters

Persistency and storage

- Sessions
- Cookies
- File storage

Day 2 - OOP and database

The essence of Object Oriented Programming

- Basics
- Properties
- *Constructor* and *destructor*
- Keyword *$this*
- Visibility
- Constants
- Inheritance
- *Abstract*, *Final* and *Static*
- Magic methods
- Errors and exceptions
- HTTP headers
- Interfaces and traits
- Namespaces

Database

- Mysqli
- PDO

Day 3 - External libraries

External libraries

- Autoload
- Composer
- Dotenv
- PHPUnit
- PHP CLI

Ecosystem

- Extensions
- Frameworks
- CMS

8. THE BASICS OF SQL



a. Description

*This hands-on training is designed to give developers, analysts, and IT professionals a solid foundation in SQL. You'll learn how to **design databases**, **write efficient queries**, and **analyze data** using real-world exercises and best practices. Whether you're just getting started or looking to deepen your skills, this course will make you fluent in relational database querying.*

b. Educational goals

- Understand the structure and logic of relational databases
- Design and create normalized database schemas
- Write basic and advanced SQL queries to retrieve and manipulate data
- Perform data aggregation and analysis using SQL functions
- Optimize queries and apply best practices for performance

c. Prerequisites

- Basic understanding of data and its organization (spreadsheets, tables)
- Familiarity with fundamental programming or logic concepts (variables, conditions)

d. Targeted audience

- Developers
- Data analysts
- System administrators

e. Program

Day 1 - SQL fundamentals

Introduction to SQL and relational databases

- Theoretical explanation of the relational model
- Database architecture and components
- Basics of SQL: syntax, structure, and execution

Database design and schema creation

- Understanding ER diagrams and normalization
- Creating tables: *CREATE TABLE*, constraints (*PRIMARY KEY*, *FOREIGN KEY*, etc.)
- Data types and indexing

Basic SQL queries

- Retrieving data with *SELECT*, filtering with *WHERE*
- Using operators (*AND, OR, LIKE, IN*)
- Sorting results with *ORDER BY*

Day 2 - Advanced SQL concepts

Advanced queries and joins

- Understanding *INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN*
- Combining results with *UNION* and *INTERSECT*

Aggregations and subqueries

- Grouping data with *GROUP BY*, filtering with *HAVING*
- Using aggregate functions (*COUNT, SUM, AVG*, etc.)
- Writing subqueries with *EXISTS, NOT EXISTS*

Window functions

- Introduction to window functions (*OVER, PARTITION BY*)

Performance optimization

- Optimizing queries with indexing
- Common pitfalls and best practices

9. INTRODUCTION TO BASH SCRIPTING



a. Description

*This 3-day hands-on course introduces participants to Bash scripting for task automation and system administration. From writing **simple scripts** to **managing processes**, **securing code**, and **working over SSH**, you'll learn practical skills to boost productivity and control over your Linux environment. Perfect for **devs** and **sysadmins**.*

b. Educational goals

- Understand the fundamentals of the Bash shell and scripting syntax
- Write and execute basic to intermediate Bash scripts
- Automate common system tasks using loops, conditions, and variables
- Manage processes and schedule tasks
- Handle user input, signals, and logging in scripts

c. Prerequisites

- Being comfortable with CLI mode
- Familiarity with basic Linux commands

d. Targeted audience

- System administrators
- Dev(Sec)Ops engineers
- Anyone interested in automation

e. Program

Day 1 - Introduction to Bash and file manipulation

Introduction to Bash

- What is Bash?
- Why learn Bash?
- Basic commands

Writing and Running Bash scripts

- What is a Bash script?
- Create and execute a script
- The *Shebang*
- Grant permissions
- Run a script

Variables

- Understanding variables
- Declaring and using variables
- Types of variables

Conditional statements

- What are conditional statements?
- *if-else* statements
- *case* statements

Loops (*for, while, until*)

- *for* loops
- *while* loops
- *until* loops

Functions in Bash

- Why use functions?
- Defining and calling functions

Day 2 - Control structures, functions, and advanced scripting

Handling user input and arguments

- Reading user input
- Script arguments

Redirection and pipelines

- Output and error redirection
- Using pipelines

File and directory operations

- Creating and manipulating files/directories

Permissions and access rights

- Understanding Linux file permissions
- View and change permissions

Debugging and error handling

- Debugging
- Handling errors
- Handle script interruptions

Day 3 - Process management, automation, and security

Process management in Bash

- Understanding processes in Linux
- Viewing processes

Scheduling tasks

- Automating tasks
- *cron* job format

Logging and monitoring scripts

- Logging
- Checking system logs

Handling signals and traps

- What are signals?
- How to handle signals

Securing Bash scripts

- File permissions and execution restrictions
- Validating user input
- Using *sudo* securely

Working with remote servers and SSH

- Connect to a remote server
- Copying files securely
- Automating SSH tasks